WORKED SOLUTION

STEP II 2009, Question 3

Pure • Trigonometric Identities and Surds

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q3

The key idea

The identity $(*)$ turns a tangent you know at an angle x into a tangent at the angle $\frac{1}{4}\pi - \frac{1}{2}x$, which is roughly half as big. So it acts as a half-angle machine. The work is in choosing a starting angle x that lands exactly on the angle you want. After that, the job is careful surd arithmetic: rationalising, expanding and squaring.

Solution

Proving $(*)$

Write $\theta = \frac{1}{4}\pi - \frac{1}{2}x$, so that $x = \frac{1}{2}\pi - 2\theta$. Then

$$\sin x = \cos 2\theta = 1 - 2\sin^2\theta, \qquad \cos x = \sin 2\theta = 2\sin\theta\cos\theta.$$

Hence, wherever both sides are defined (that is, $\cos x \neq 0$),

$$\sec x - \tan x = \frac{1-\sin x}{\cos x} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right).$$

Cancelling $\sin\theta$ is allowed because $\cos x = 2\sin\theta\cos\theta \neq 0$ forces $\sin\theta \neq 0$. This proves $(*)$.

Part (i)

We want $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{8}\pi$, which gives $x = \frac{1}{4}\pi$. Since $\sec\frac{1}{4}\pi = \sqrt{2}$ and $\tan\frac{1}{4}\pi = 1$,

$$\tan\tfrac{1}{8}\pi = \sqrt{2} - 1.$$

For the second result, notice that $\frac{11}{24}\pi = \frac{3}{24}\pi + \frac{8}{24}\pi = \frac{1}{8}\pi + \frac{1}{3}\pi$, and we know $\tan\frac{1}{3}\pi = \sqrt{3}$. The addition formula gives

$$\tan\tfrac{11}{24}\pi = \frac{\tan\frac{1}{8}\pi + \tan\frac{1}{3}\pi}{1 - \tan\frac{1}{8}\pi\tan\frac{1}{3}\pi} = \frac{(\sqrt{2}-1)+\sqrt{3}}{1-(\sqrt{2}-1)\sqrt{3}} = \frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1},$$

which is the required result.

Part (ii)

Rather than rationalise a denominator with three terms, it is quicker to show that

$$(2+\sqrt{2}+\sqrt{3}+\sqrt{6})(\sqrt{3}-\sqrt{6}+1) = \sqrt{3}+\sqrt{2}-1.$$

Multiplying $\sqrt{3}-\sqrt{6}+1$ by each term of the first bracket in turn, using $\sqrt{12}=2\sqrt{3}$ and $\sqrt{18}=3\sqrt{2}$:

$$\begin{aligned}
2(\sqrt{3}-\sqrt{6}+1) &= 2\sqrt{3}-2\sqrt{6}+2,\\
\sqrt{2}(\sqrt{3}-\sqrt{6}+1) &= \sqrt{6}-2\sqrt{3}+\sqrt{2},\\
\sqrt{3}(\sqrt{3}-\sqrt{6}+1) &= 3-3\sqrt{2}+\sqrt{3},\\
\sqrt{6}(\sqrt{3}-\sqrt{6}+1) &= 3\sqrt{2}-6+\sqrt{6}.
\end{aligned}$$

Now we collect terms. The constants give $2+3-6=-1$. The $\sqrt{2}$ terms give $1-3+3=1$. The $\sqrt{3}$ terms give $2-2+1=1$. The $\sqrt{6}$ terms give $-2+1+1=0$. The total is $\sqrt{3}+\sqrt{2}-1$. The denominator $\sqrt{3}-\sqrt{6}+1 \approx 0.28$ is non-zero, so we may divide by it, and

$$\frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1} = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}.$$

Ask yourself Which value of x should go into $(*)$ for the final part?

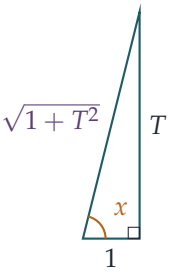
We need $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{48}\pi$, so $\frac{1}{2}x = \frac{11}{48}\pi$ and $x = \frac{11}{24}\pi$. This is exactly the angle whose tangent we found in (i) and (ii).

Part (iii)

Let $T = \tan\frac{11}{24}\pi = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}$. By $(*)$ with $x=\frac{11}{24}\pi$,

$$\tan\tfrac{1}{48}\pi = \sec\tfrac{11}{24}\pi - T.$$

For the secant, $\sec^2 x = 1+\tan^2 x$. Since $0 < \frac{11}{24}\pi < \frac{1}{2}\pi$, the cosine is positive, and so we take the positive root, $\sec\frac{11}{24}\pi = \sqrt{1+T^2}$. The triangle below shows the same thing.



Squaring T, the squares of the four terms give $4+2+3+6=15$. The cross terms give

$$2(2\sqrt{2}+2\sqrt{3}+2\sqrt{6}+\sqrt{6}+\sqrt{12}+\sqrt{18}) = 2(5\sqrt{2}+4\sqrt{3}+3\sqrt{6}) = 10\sqrt{2}+8\sqrt{3}+6\sqrt{6}.$$

So $1 + T^2 = 16 + 10\sqrt{2} + 8\sqrt{3} + 6\sqrt{6}$, and therefore

$$\tan \tfrac{1}{48}\pi = \sqrt{16 + 10\sqrt{2} + 8\sqrt{3} + 6\sqrt{6}} - 2 - \sqrt{2} - \sqrt{3} - \sqrt{6},$$

as required.

Checks

- $\sqrt{2} - 1 \approx 0.41421$, which matches $\tan 22.5°$.
- $2 + \sqrt{2} + \sqrt{3} + \sqrt{6} \approx 7.59575$, which matches $\tan 82.5°$. The fraction in (i) gives the same value.
- The final expression is $\sqrt{57.45\ldots} - 7.59575 \approx 0.0655435$, which matches $\tan 3.75° \approx 0.0655435$.
- A quick test of $(*)$: at $x = 0$ both sides equal 1, and at $x = \frac{1}{3}\pi$ both sides equal $2 - \sqrt{3} = \tan \frac{1}{12}\pi$.

Common slips

- Solving $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{48}\pi$ wrongly and using $x = \frac{11}{48}\pi$. The $\frac{1}{2}$ must be undone.
- Missing cross terms when squaring a four-term surd. There are six of them, each doubled.
- Taking $\sec x = \sqrt{1 + \tan^2 x}$ without saying why the positive root is the right one.

Beyond STEP

Repeatedly halving angles in this way is how exact values such as $\tan \frac{\pi}{48}$, nested square roots and all, are built up. It is the algebraic side of the classical question of which regular polygons can be constructed with ruler and compass. Gauss showed that the answer depends on $\cos \frac{2\pi}{n}$ being expressible using only square roots, and Galois theory explains why. For a first look at fields and field extensions, which is the language of this story, try Allenby, *Rings, Fields and Groups*.

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